Timothy Shen

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Education

University of California, Los Angeles **September 2024 – Present**
Ph.D. Student, Biostatistics *Los Angeles, CA*
Advisors: Dr. Christina Ramirez and Dr. Brunilda Balliu

University of California, Davis **March 2024**
B.S., Statistics *Davis, CA*
Minors: Computer Science and Mathematics

Research Areas

Statistics: Machine Learning (Tree-Based Models, Feature Selection, Deep Learning), Longitudinal Data, High-Dimensional Data, Unsupervised Learning, Network Analysis
Applied Research Areas: HIV and Aging, Epigenomics, Neuroscience, Wearable Devices and Digital Mental Health
Computational Areas: R (R Package Development, R Shiny), Python, TensorFlow, SQL, SAS, MATLAB, High-Performance Computing (SLURM, SGE), Bash, Git/GitHub, Docker

Research Experience

Graduate Student Researcher **January 2026 – Present**
Digital Mental Health Study, UCLA Department of Computational Medicine *Los Angeles, CA*

- Developing machine learning (ML) and deep learning models that use passive wearable and smartphone sensing temporal data to predict depression severity, under applications of both screening and symptom monitoring.
- Addressing the complexity of wearable data (high-dimensional, multimodal, irregularly sampled, and structural missingness) through data quality control and feature engineering of passive sensing signals.
- Implementing deep learning imputation of missing sensor data with autoencoders and self-attention transformers to reproduce observed missingness patterns for reasonable training and evaluation.
- Using learned representations from these models as inputs to downstream ML prediction models, extending representation learning methods to improve depression prediction.
- Performing subtyping to characterize heterogeneity in depression, uncovering latent mental health states and their transitions with a multivariate hidden Markov model (mHMM) and relating them to wearable derived digital markers.

Supervised by Dr. Brunilda Balliu, Assistant Professor of Computational Medicine, UCLA Geffen School of Medicine

Graduate Student Researcher **August 2025 – Present**
Aging Adult Brain Connectome Study, Semel Institute for Neuroscience and Human Behavior *Los Angeles, CA*

- Applying network-guided machine learning techniques to neuroscience data from the Aging Adult Brain Connectome, including brain markers, non-behavioral features, and biomarkers, to model cognitive outcomes.
- Characterizing how age modifies the correlation network structure of brain and biomarker data by comparing correlation networks across different aging groups.
- Evaluating how age interacts with brain, biomarker, and other features to change prediction of cognitive performance, quantifying age-dependent differences in predictor importance.

Supervised by Dr. Helen Lavretsky, Professor of Psychiatry and Dr. Christina Ramirez, Professor of Biostatistics, UCLA

Graduate Student Researcher **August 2024 – Present**
UCLA Fielding School of Public Health *Los Angeles, CA*

- Analyzing 800,000+ CpG sites of DNA methylation to identify HIV-related epigenetic biomarkers, contributing to early detection and aging research.
- Designing mossyforest, a novel machine learning R package for interpretable, network-aware, tree-based feature selection on highly correlated, high-dimensional data, using SHapley Additive exPlanations (SHAP) and Recursive Feature Elimination.
- Implementing cluster simulations to evaluate the feature selection method against artificial $p \gg n$ datasets.
- Extending the method to high-dimensional, correlated longitudinal data through time-aware network reduction and mixed-effects trees.

Supervised by Dr. Christina Ramirez, Professor of Biostatistics, UCLA Fielding School of Public Health

Graduate Student Researcher **June 2025 – September 2025**
Screening & Treatment for Anxiety & Depression, UCLA Department of Comp. Medicine *Los Angeles, CA*

- Applied an unsupervised statistical learning method, a multivariate hidden Markov model (MHMM), to longitudinal digital phenotyping data from wearable devices to uncover latent mental health states, including depression, anxiety, suicidality, and mania.
- Engineered and analyzed passive sensing data (e.g., sleep, activity, heart rate variability) to investigate associations with mental health state transitions over time.
- Identified wearable-derived digital markers significantly associated with transitions between latent states, offering insights into continuous monitoring and early detection of affective disorders.

Supervised by Dr. Brunilda Balliu, Assistant Professor of Computational Medicine, UCLA Geffen School of Medicine

Undergraduate Researcher **May 2021 – December 2022**
Stanford Genome Technology Center (SGTC) *Palo Alto, CA*

- Improved analytic software for biostatistical research teams by developing R packages and interactive R Shiny dashboards, reducing analysis time and facilitating real-time collaboration.
- Analyzed the relationship between fourteen metal types and Chronic Fatigue Syndrome (CFS) in patients versus controls.

Supervised by Dr. Wenzhong Xiao, Professor at Harvard Medical School, and Dr. Chia Jung Chang, Professor at SGTC

Research Assistant **August 2019 – September 2020**
Harvard Medical School and Massachusetts General Hospital (MGH) *Boston, MA*

- Processed 100+ files of medical research data (de-identification, scientific data formatting, and data sharing) and unified data entries through REDCap and Excel.
- Tested questionnaires and surveys for Chronic Fatigue Syndrome patients to improve ease of use for patients and researchers.
- Designed a website showcasing all seven research projects to improve communication among researchers.

Supervised by Dr. Wenzhong Xiao, Professor at Harvard Medical School, and Dr. Li-Yuan Hung, Research Scientist at MGH

Industry Experience

Statistics Research Intern **June 2023 – December 2024**
Immucor, Inc. *Mountain View, CA*

- Developed a detection model for Human Leukocyte Antigen (HLA) loss to identify post–stem cell transplant relapse events, improving early detection accuracy.
- Built reproducible R pipelines to process next-generation sequencing (NGS) data and automate HLA loss detection.
- Led weekly research presentations to the Biostatistics Division, communicating clinical relevance to support continuation of the project.

Supervised by Dr. Ming Li, Vice President of Bioinformatics & Head of Next-Generation Sequencing (NGS) Software

Awards

Graduate Research Mentorship Fellowship | University of California, Los Angeles **2026**
Division of Graduate Education, UCLA

- Awarded to develop scalable and interpretable longitudinal Random Forest models as part of my dissertation research.
- Received a $30,000 fellowship supporting research throughout the 2026–2027 academic year.

Graduate Summer Research Mentorship Fellowship | University of California, Los Angeles **2026**
Division of Graduate Education, UCLA

- Awarded to build a network-aware Random Forest feature selection method incorporating stability selection for high-dimensional correlated data.
- Received a $6,000 fellowship supporting research throughout summer 2026.

Graduate Research Mentorship Fellowship | University of California, Los Angeles **2025**
Division of Graduate Education, UCLA

- Awarded to conduct HIV research identifying epigenome-wide association study (EWAS) CpG sites associated with mechanisms of aging, using machine learning feature selection.
- Received a $30,000 fellowship supporting research throughout the 2025–2026 academic year.

Competitive Edge Fellow | University of California, Los Angeles **2024**

Division of Graduate Education, UCLA

- Selected among incoming UCLA Ph.D. students to begin research early in the summer prior to the first academic quarter in developing network-aware feature selection methods through the use of Shapley values.
- Received a $6,000 stipend supporting research throughout summer 2024.

Prized Writing | University of California, Davis **2023**

"Empowering the Underprivileged: How Machine Learning is Revolutionizing Humanitarian Assistance in Low-Income Countries"

- Annual juried competition recognizing outstanding undergraduate writing at UC Davis.
- Selects around 20 pieces from over 400 entries by undergraduate students in non-creative writing courses.

Academic Service & Leadership

Treasurer **2026**

Biostatistics Student Association, UCLA *Los Angeles, CA*

- Manage the association's budget, funding requests, and event expenditures.

Graduate Student Representative, Transportation Service Advisory Board **2025**

Graduate Student Association, UCLA *Los Angeles, CA*

- Advocated for 14,000+ graduate students in advising UCLA Transportation on campus transit, parking, and commuting services.

Board Member **2024**

Biostatistics Student Association, UCLA *Los Angeles, CA*

- Helped organize academic and community events for the biostatistics graduate student community.

Teaching Experience

Teaching Assistant, BIOSTAT 100: Introduction to Biostatistics **2025**

UCLA Fielding School of Public Health *Los Angeles, CA*

- Led weekly labs and hosted office hours for undergraduate students, covering the basics of R and foundational biostatistics.

Teaching Assistant, BIOSTAT 201A/B: Introduction to Biostatistics **2024**

UCLA Fielding School of Public Health *Los Angeles, CA*

- Led discussion sections and hosted office hours for graduate students in a two-course sequence on the foundations of biostatistics, regression, and inference for public health students.

Publications

Timothy Shen. "Empowering the Underprivileged: How Machine Learning is Revolutionizing Humanitarian Assistance in Low-Income Countries". In: *Prized Writing, UC Davis* (2023).

Software

Timothy Shen and Christina Ramirez. *mossyforest: Network Aware Stability Selection with Random Forests*. R package version 0.1.0. 2026. URL: https://github.com/timothyzshen/mossyforest.